Faà di Bruno, Leibniz Rules, and Matrix Multiplication

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§1.1 Motivation

This note jumps among several topics: high-order derivatives of composite functions, Leibniz-type determinant differentiation, and the beginning of matrix multiplication. The unifying theme is that complicated formulas become manageable once the indexing convention is chosen correctly.

§1.2 Faà di Bruno formula

For one-variable functions, the n-th derivative of a composite $f \circ g$ is governed by Faà di Bruno's formula:

$$(f \circ g)^{(n)}(x) = \sum_{m=1}^{n} \sum_{(k_1,\ldots,k_n)\in\Gamma_{m,n}} \frac{n!}{k_1! \cdots k_n!} f^{(m)}(g(x)) \prod_{j=1}^{n} \left(\frac{g^{(j)}(x)}{j!} \right)^{k_j},$$

where

$$\Gamma_{m,n} = \{(k_1, \ldots, k_n) : k_j \in \mathbb{Z}_{\geq 0}, k_1 + \cdots + k_n = m, k_1 + 2k_2 + \cdots + nk_n = n\}.$$

This is the high-order chain rule. The indexing records how the n derivatives are partitioned among the derivatives of g.

§1.3 Leibniz's differential notation

If $y = f(x)$, then

$$dy = f'(x)\, dx.$$

Taking another differential gives

$$d^2y = d(f'(x)dx) = f''(x)dx^2 + f'(x)d^2x.$$

If $x = x(t)$, this corresponds to the ordinary chain rule

$$\frac{dy}{dt} = f'(x(t))x'(t).$$

For the third differential, one obtains

$$d^3y = f'''(x)dx^3 + 3f''(x)dx\, d^2x + f'(x)d^3x.$$

This is already a visible shadow of Faà di Bruno.

§1.4 Parametric derivatives

If a curve is given parametrically by

$$x = x(t), \qquad y = y(t),$$

then

$$y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

The second derivative is

$$y'' = \frac{x'(t)y''(t) - y'(t)x''(t)}{(x'(t))^3}.$$

This can be written as a determinant:

$$y'' = \frac{\begin{vmatrix} x' & y' \\ x'' & y'' \end{vmatrix}}{(x')^3}.$$

§1.5 Differentiating determinants

For an $n \times n$ matrix of functions $A(x) = (f_{ij}(x))$, the derivative of the determinant is obtained by differentiating one row at a time:

$$\frac{d}{dx} \det A(x) = \sum_{i=1}^{n} \det \begin{pmatrix} f_{11} & \cdots & f_{1n} \\ \vdots & & \vdots \\ f'_{i1} & \cdots & f'_{in} \\ \vdots & & \vdots \\ f_{n1} & \cdots & f_{nn} \end{pmatrix}.$$

This is simply the multilinearity of the determinant.

§1.6 Matrix multiplication

If

$$A \in M_{m\times n}, \qquad B \in M_{n\times p},$$

then their product $C = AB \in M_{m\times p}$ has entries

$$C_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

The note emphasizes that in general $AB \neq BA$, while associativity still holds:

$$A(BC) = (AB)C.$$

For block matrices,

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}.$$

§1.7 Schur complement

The block elimination identity in the note is

$$\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}.$$

The matrix

$$D - CA^{-1}B$$

is the Schur complement of A in the block matrix. It is the block-matrix version of Gaussian elimination.

§1.8 What the pattern suggests

The note moves from high-order calculus to matrix algebra, but the same mathematical taste appears in both places: the right notation makes the formula intelligible. Multi-indices organize Faà di Bruno, determinants organize parametric derivatives, and block matrices organize elimination.

§1.9 Faà di Bruno as chain rule for higher jets

The ordinary chain rule controls first derivatives of a composition. Faà di Bruno controls all higher derivatives and is naturally expressed through partitions:

$$\frac{d^n}{dx^n} f(g(x)) = \sum f^{(k)}(g(x)) \prod_j \left(\frac{g^{(j)}(x)}{j!} \right)^{m_j} \frac{n!}{\prod_j m_j!}.$$

The combinatorics records how n differentiations are distributed among the inner function.

§1.10 Leibniz rules and derivations

A derivation D satisfies

$$D(ab) = D(a)b + aD(b).$$

Higher Leibniz formulas are binomial expansions of repeated derivations:

$$D^n(ab) = \sum_{k=0}^{n} \binom{n}{k} D^k(a) D^{n-k}(b).$$

§1.11 Matrix multiplication as composition

Matrix multiplication is the coordinate expression of composing linear maps. The index formula

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}$$

contracts the output coordinate of B with the input coordinate of A.

§1.12 Schur complements

For a block matrix, the Schur complement

$$D - CA^{-1}B$$

records what remains after eliminating the variables controlled by A. It is block Gaussian elimination, and it appears in determinants, inverses, covariance matrices, and constrained quadratic forms.

§1.13 Composition, derivations, and elimination

The note links differentiation and linear algebra through composition. Faà di Bruno is higher chain rule, Leibniz is derivation algebra, matrix multiplication is composition, and Schur complements are elimination in block form.